

High Performance and Distributed Computing for Big Data

Unit 4: AWS - S3

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- AWS S3: Storing files in the cloud
 - Installing AWS CLI
 - Configuring AWS credentials
 - Creating an S3 bucket
 - Syncing a local directory to upload to a bucket
 - Loading files from the bucket to the notebook from python
 - Writing files from the notebook to the bucket from python
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S3 - Storing files in the cloud

What is S3?

Amazon S3 (Simple Storage Service) is an object storage service that offers industry-leading scalability, data availability, security, and performance. It is designed to store and retrieve any amount of data from anywhere on the web.



S3 - Buckets and Objects

- **Buckets:** In Amazon S3, a bucket is a unique container for objects. Every object is stored within a bucket.
 - The bucket name must be globally unique across all existing bucket names in Amazon S3.
 - Buckets are used to store and organize objects.
- **Objects:** Objects are the fundamental entities within a bucket. They consist of data and metadata.
 - The information within an object is stored as a **key-value** pair.
 - The key, which is a unique identifier, is used to organize objects within the bucket. It is often formatted as a prefix to the object name.

For example, we can create a bucket called `hdc-b-{your-name}` and store objects organized by session or other criteria.

```
data-{your-name}/  
  project1/data.csv  
  project2/data.csv
```

Terms

- **Key** = *prefix + object name*
 - **prefix** = project1/ or project2/
 - **object name** = data.csv

Amazon S3 pricing is based on five factors:

1. **Storage Class:** The cost depends on the storage class used (Standard, Intelligent-Tiering, One Zone-IA, etc.).
2. **Storage:** The total volume of data stored per month.
3. **Requests:** The number and type of requests made.
4. **Data Transfer:** The cost of transferring data can vary by region and is also affected by whether data is transferred in or out.
5. **Management & Replication:** Additional features like data replication or management operations can also affect the cost.

For detailed information, you can refer to the [Amazon S3 Pricing Page](#).

AWS CLI

What is the AWS CLI?

The AWS CLI is a tool that allows you to interact with AWS services from the command line. It is a powerful tool that can be used to automate tasks and manage your AWS resources.

```
[ec2-user@ip-172-31-86-82 ~]$ aws
```

```
usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
```

To see help text, you can run:

```
aws help
```

```
aws <command> help
```

```
aws <command> <subcommand> help
```

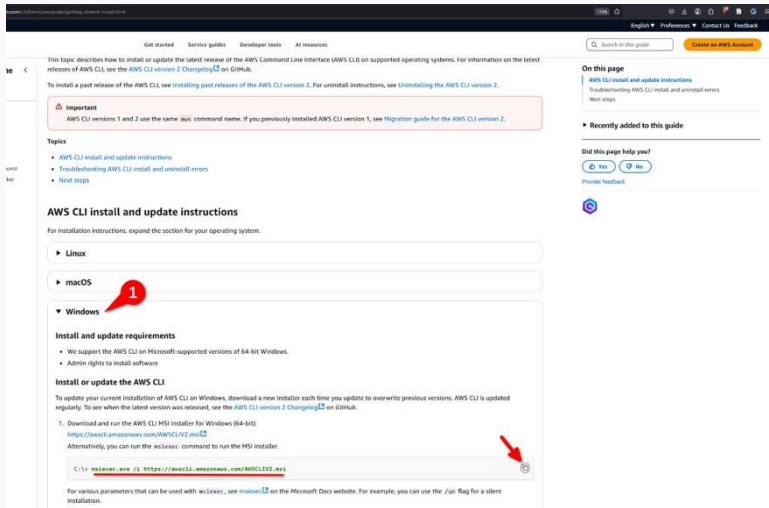
```
aws: error: the following arguments are required: command
```

```
[ec2-user@ip-172-31-86-82 ~]$
```

This is the base command which by itself doesn't do anything. What we are going to do now is to configure the AWS CLI with our credentials so we can later run commands that can access other AWS resources like S3.

Installing the AWS CLI

Visit the [AWS CLI installation guide](#) to install the AWS CLI on your machine.



The screenshot shows the AWS CLI installation guide page. The page is titled "AWS CLI install and update instructions". It includes a sidebar with navigation links and a main content area. The "Windows" section is expanded, showing "Install and update requirements" and "Install or update the AWS CLI". A red circle with the number "1" highlights the "Windows" section header. A red arrow points to the "msiexec" command in the "Install or update the AWS CLI" section.

Important
AWS CLI versions 1 and 2 use the same `aws` command name. If you previously installed AWS CLI version 1, see [Migration guide for the AWS CLI version 2](#).

Topics

- [AWS CLI install and update instructions](#)
- [Troubleshooting AWS CLI install and uninstall errors](#)
- [Next steps](#)

AWS CLI install and update instructions

For installation instructions, expand the section for your operating system.

- ▶ Linux
- ▶ macOS
- ▼ Windows

Install and update requirements

- We support the AWS CLI on Microsoft-supported versions of 64-bit Windows.
- Admin rights to install software

Install or update the AWS CLI

To update your current installation of AWS CLI on Windows, download a new installer each time you update to overwrite previous versions. AWS CLI is updated regularly. To see when the latest version was released, see [AWS CLI version 2 Changelog](#) on GitHub.

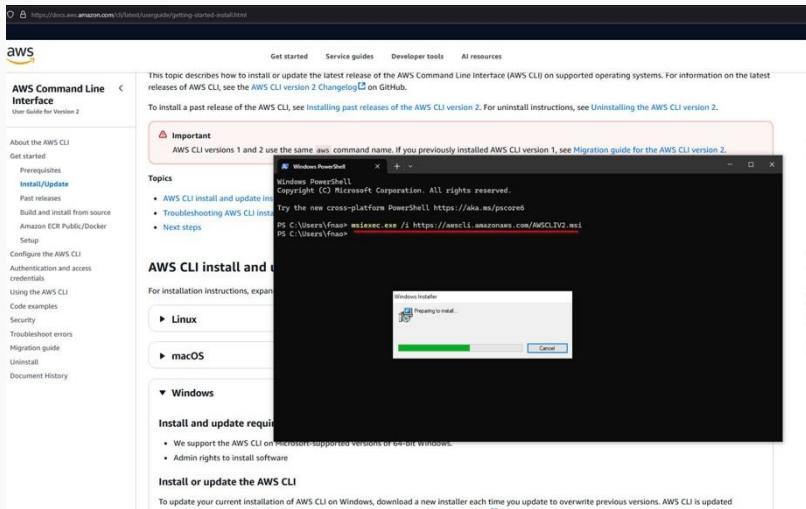
1. Download and run the AWS CLI MSI installer for Windows (64-bit):
<https://awscli.amazonaws.com/AWSCLIV2.msi>
Alternatively, you can run the `msiexec` command to run the MSI installer:

```
C:\> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
```

For various parameters that can be used with `msiexec`, see [msiexec](#) on the Microsoft Docs website. For example, you can use the `/qn` flag for a silent installation.

Installing the AWS CLI

Paste the command on the terminal and wait for the installation to complete.



The screenshot displays the AWS Command Line Interface (CLI) user guide for Version 2 on the AWS website. The page includes a navigation sidebar on the left with links for 'About the AWS CLI', 'Get started', 'Prerequisites', 'Install/Update', 'Past releases', 'Build and install from source', 'Amazon ECR Public/Docker', 'Setup', 'Configure the AWS CLI', 'Authentication and access credentials', 'Using the AWS CLI', 'Code examples', 'Security', 'Troubleshoot errors', 'Migration guide', 'Uninstall', and 'Document History'. The main content area features an 'Important' note stating that AWS CLI versions 1 and 2 use the same 'aws' command name, and a 'Topics' section with links for 'AWS CLI install and update instructions', 'Troubleshooting AWS CLI installation', and 'Next steps'. Below this, there are sections for 'AWS CLI install and update instructions' for Linux, macOS, and Windows. The Windows section includes a note about supported versions and admin rights, and a section for 'Install or update the AWS CLI' which instructs users to download a new installer to overwrite previous versions.

Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\fnaao> msiexec.exe /i https://awscli.amazonaws.com/AWSCLIV2.msi
PS C:\Users\fnaao>
```

Windows Installer

Preparing to install...

Cancel

Installing the AWS CLI

Or if you are on MacOS, go to the MacOS section and also paste the corresponding commands on your terminal.

The screenshot shows the AWS Command Line Interface User Guide for Version 2. The left sidebar contains a navigation menu with links like 'About the AWS CLI', 'Get started', 'Prerequisites', 'Install/Update', 'Past releases', 'Build and install from source', 'Amazon ECR Public/Docker', 'Setup', 'Configure the AWS CLI', 'Authentication and access credentials', 'Using the AWS CLI', 'Code examples', 'Security', 'Troubleshoot errors', 'Migration guide', 'Uninstall', and 'Document History'. The main content area is titled 'macOS' and includes a section 'Install and update requirements' with two bullet points. Below this is a 'macOS version support matrix' table. Further down is a section 'Install or update the AWS CLI' with a paragraph and a code block containing two terminal commands. A red callout '1' points to the 'macOS' section header, and a red callout '2' points to the 'Command line installer - All users' tab. A red arrow points to a copy icon next to the terminal commands.

macOS

Install and update requirements

- We support the AWS CLI on macOS versions 11 and later. For more information, see [macOS support policy updates for the AWS CLI v2](#) on the AWS Developer Tools Blog.
- Because AWS doesn't maintain third-party repositories, we can't guarantee that they contain the latest version of the AWS CLI.

macOS version support matrix

AWS CLI version	Supported macOS version
2.21.0 – current	11+
2.17.0 – 2.20.0	10.15+
2.0.0 – 2.16.12	10.14 and below

Install or update the AWS CLI

If you are updating to the latest version, use the same installation method that you used in your current version. You can install the AWS CLI on macOS in the following ways.

Command line installer - All users

If you have `sudo` permissions, you can install the AWS CLI for all users on the computer. We provide the steps in one easy to copy and paste group. See the descriptions of each line in the following steps.

```
$ curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
$ sudo installer -pkg AWSCLIV2.pkg -target /
```

Guided installation instructions

- Download the file using the `curl` command. The `-o` option specifies the file name that the downloaded package is written to. In this example, the file is written to `AWSCLIV2.pkg` in the current folder.

Configuring the AWS Credentials

We're now going to visit the Learner Lab page on the AWS Academy website to get our credentials. You have [this guide](#) and [this guide](#) available on the subject's [website](#) to help you with setting up AWS. Wait until the lab loads and you see the page below.

The screenshot shows the AWS Academy Learner Lab interface. The browser address bar displays `awsacademy.instructure.com/courses/111255/modules/items/70403094`. The breadcrumb navigation shows `ALLv2EN-... > Modules > AWS Acad... > Launch AWS Academy Learner Lab`. The interface includes a left sidebar with navigation links: Home, Modules, Discussions, Grades, Lucid (Whiteboard), Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area features a terminal window with the prompt `eee_w_4181718@runweb163543:~$` and a status bar indicating `AWS` and `Used $1.5 of $50`. A top bar shows the lab duration `01:03` and buttons for `Start Lab`, `End Lab`, `AWS Details`, `Readme`, and `Reset`. On the right, the **Learner Lab** section lists several links: [Environment Overview](#), [Environment Navigation](#), [Access the AWS Management Console](#), [Region restriction](#), [Service usage and other restrictions](#), [Using the terminal in the browser](#), [Running AWS CLI commands](#), [Using the AWS SDK for Python](#), [Preserving your budget](#), [Accessing EC2 Instances](#), [SSH Access to EC2 Instances](#), [SSH Access from Windows](#), and [SSH Access from a Mac](#).

Configuring the AWS Credentials

Now click on **AWS Details** and then on **Show** to reveal your credentials.

The screenshot shows the AWS Academy Learner Lab interface. The breadcrumb navigation at the top reads: **ALLv2EN-...** > **Modules** > **AWS Acad...** > **Launch AWS Academy Learner Lab**. The main header bar displays the **AWS** logo, **Used \$1.6 of \$50**, a timer at **01:01**, and buttons for **Start Lab**, **End Lab**, **AWS Details** (highlighted with a red circle 1), **Readme**, **Reset**, and a close icon. On the left sidebar, the **Modules** section is active, with sub-links for **Home**, **Discussions**, **Grades**, and **Lucid (Whiteboard)**. The central area contains a terminal window with the prompt `eee_w_4181718@runweb163552:~$`. On the right, the **Cloud Access** panel is open, showing **AWS CLI:** with a **Show** button (highlighted with a red circle 2). Below this, the **Cloud Labs** section displays session information: **Remaining session time: 01:00:53(61 minutes)**, **Session started at: 2025-03-03T08:48:13-0800**, **Session to end at: 2025-03-03T12:48:13-0800**, and **Accumulated lab time: 1 day 05:41:00 (1781 minutes)**.

Configuring the AWS Credentials

You'll see some text containing your credentials.

Used \$2.1 of \$50 03:46 ▶ Start Lab ■ End Lab ⓘ AWS Details ⓘ Readme ↺ Reset ✕

Cloud Access Close

AWS CLI:
Copy and paste the following into ~/.aws/credentials

```
[default]
aws_access_key_id=ASIA2CKYVHJA03I2XRZ3
aws_secret_access_key=dudm/D3hbusUkb6iqzEXFcVXjjQkghDy3+AL
EOa3
aws_session_token=IQoJb3JpZ2luX2VjELf////////wEaCXVzLXd1
c3QtMiJGMEQCIE00uLiVzFVLHv5EbZPZj8hPqPJGou84lGfM0ezfiSP0Ai
B+S2vrdPaz4TFMEYXwe1e0Hr/C05m0B3N72MZHQ1n0DyqzAgj
w////////8BEAEaDDY5MjIxMjU0NjExMiIMCDFeGSKhijJKhgHCKocCd
6djgX83VD2PG83XsoMK972gcLP6P8T0ZvujiY4RT0S2uqbjAW/2cNwdpuk
yH8LX0zAjI00CzPwr+M5HrC/7vUPZSANonVQdJvmbBrXZ+ln8SzzTuBFq1
zD6bRhnV7iHhA8naNoXe4eNAM0C09HQ6JmfVAoPTQ3nGDPGKEfIPmG6KzI
R3H0vmuZSvsRPb4MctQxk/x4m1d4KuJ6lDeEWobNnd7fq1kvFfi+C/Gvc5
```

Configuring the AWS Credentials

Next we need to paste that text to a file called `credentials` inside the `.aws` folder in our home directory.

Make sure the `.aws` folder exists on your local machine by running `mkdir .aws` command (remember if it throws an error there's nothing to worry about, it just means the folder already exists). Now we are going to create the `credentials` file inside the `.aws` folder. Run the following command:

```
notepad .aws/credentials.
```

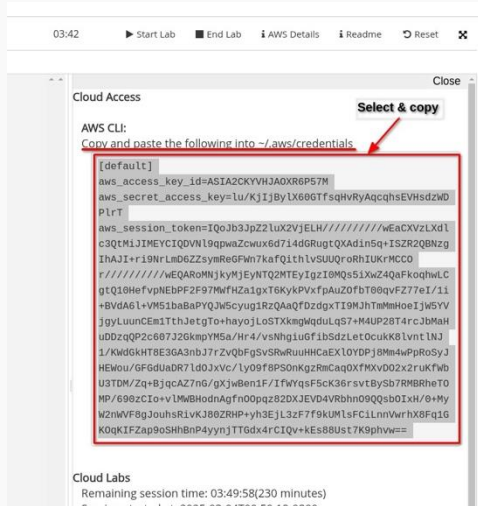
or for MacOS users:

```
open .aws/credentials.
```

This will open a text editor where you can write the credentials.

Configuring the AWS Credentials

Go back to the AWS Academy website and copy the text containing your credentials.



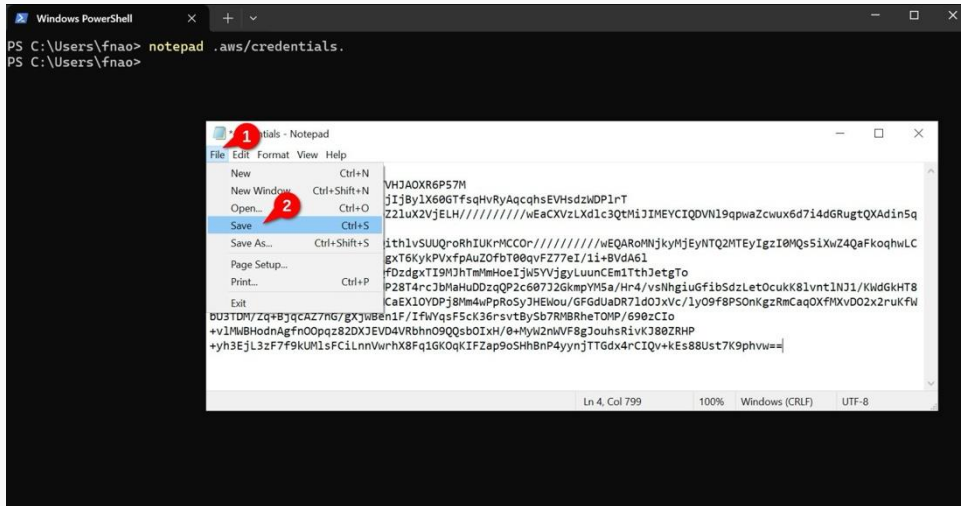
The screenshot shows a web interface for 'Cloud Access' with a 'Close' button in the top right. Below the title, there are navigation links: 'Start Lab', 'End Lab', 'AWS Details', 'Readme', 'Reset', and a close icon. The main content area is titled 'AWS CLI:' and contains the instruction 'Copy and paste the following into ~/.aws/credentials'. A red box highlights the following text, and a red arrow points to it from the 'Select & copy' button located above the box:

```
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjByLx60GTfsqHvRyAqcqhsEVHsdzWD
p1rT
aws_session_token=IQoJb3JpZ2luX2VjELH////////wEaCXVzLXdl
c3QtMiJIMEYCIQDVNl9qpwaZcwux6d7i4dGRugtQXAdin5q+ISZR2QBnzg
IhAJI+ri9NrLmd6ZZsymReGFwn7kaFqithlvSUUQroRhiUKrMCCO
r////////wEQARoMNjkyMjEyNTQ2MTEyIgzI0MQs5iXwZ4QaFkoqhWLC
gtQ10HefvpNEbPF2F97MwfHza1gxT6KykPVxfpAuZ0fbT00qvFZ77eI/i
+BVdA6l+VM51baBaPYQJw5cyug1RzQAaQfDzdgxTI9MJhTmMmHoeIjW5YV
jgyLuunCEm1TthJettTo+hayojLoSTXkmgWqduLqS7+M4UP28T4rcJbMaH
uDDzqQP2c607J2GkmpYM5a/Hr4/vsNhg1uGf1bSdzLet0cukK8lvntlNJ
1/KWdGkHT8E3GA3nbJ7rZvQbFgSvSRwRuHHHCaEXLOYDPj8Mm4wPpRoSyJ
HEWou/6FGdUaDR7ld0Jxvc/ly09f8PS0nKgZrmCaQ0XfMXvD02x2ruKfWb
U3TDM/Zq+BjqcAZ7n6/gXjwBen1F/IfwYqsF5cK36rsvtBySb7RMBRheTO
MP/690zCIo+vLMwBHodnAgfn00pqz82DXJEVD4VRbhn09QQsb0IxH/0+My
W2nWVF8gJouhsRlvKJ80ZRHP+yh3EjL3zF7f9kUmlsFC1LnnVvrhX8Fq1G
K0kQIFZap9oSHhBnP4yynjTTGdx4rCIQv+kEs88Ust7K9phvw==
```

Below the credentials, the 'Cloud Labs' section shows 'Remaining session time: 03:49:58(230 minutes)' and 'Session started at: 2025-02-04T00:00:00.000Z'.

Configuring the AWS Credentials

Now go back to the text editor. Paste the credentials and save the file. You can now exit the text editor.



Configuring the AWS Credentials

We can check the contents of the file using `cat`:

```
PS C:\Users\fnao> cat .aws\credentials
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
aws_session_token=IQoJb3JpZ2...
PS C:\Users\fnao>
```

Configuring the AWS Credentials

To test if the configuration was successful, run `aws sts get-caller-identity` and you should see something like this:

```
PS C:\Users\fnao> aws sts get-caller-identity
{
  "UserId": "AROA2CKYVHJALK46ZMHVM:user3869188=Ferran_Aran_Test",
  "Account": "692212546112",
  "Am": "am:aws:sts::692212546112:assumed-role/voclabs/user3869188=Ferran_Aran_Test"
}

PS C:\Users\fnao>
```

Great! We can now run AWS CLI commands on our local machine to manage our AWS resources. For example, we can upload files to an S3 bucket.

S3 Buckets

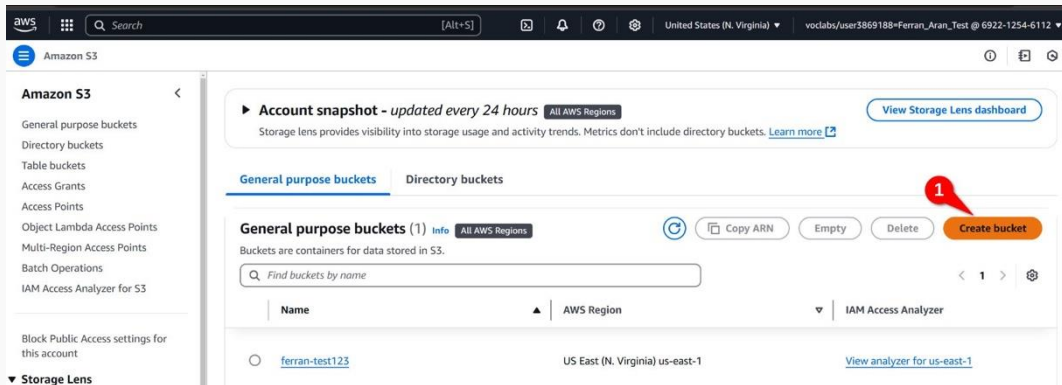
Creating a S3 bucket

Use the searchbar to head to the S3 service.



Creating a S3 bucket

Click on **Create bucket**.



The screenshot shows the Amazon S3 console interface. On the left is a navigation sidebar with the 'Amazon S3' header and a list of options: General purpose buckets, Directory buckets, Table buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, IAM Access Analyzer for S3, Block Public Access settings for this account, and Storage Lens. The main content area has a top banner for 'Account snapshot - updated every 24 hours' and a 'View Storage Lens dashboard' button. Below this are tabs for 'General purpose buckets' (selected) and 'Directory buckets'. The 'General purpose buckets' section shows 'General purpose buckets (1)' with an 'Info' link and 'All AWS Regions' filter. It includes a search bar 'Find buckets by name', a refresh button, 'Copy ARN', 'Empty', 'Delete', and a highlighted 'Create bucket' button (indicated by a red circle with the number 1). Below the buttons is a table with columns 'Name', 'AWS Region', and 'IAM Access Analyzer'. The table contains one entry: a bucket named 'ferran-test123' in the 'US East (N. Virginia) us-east-1' region, with a link to 'View analyzer for us-east-1'.

Amazon S3

General purpose buckets
Directory buckets
Table buckets
Access Grants
Access Points
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

Account snapshot - updated every 24 hours **All AWS Regions** [View Storage Lens dashboard](#)

Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets **Directory buckets**

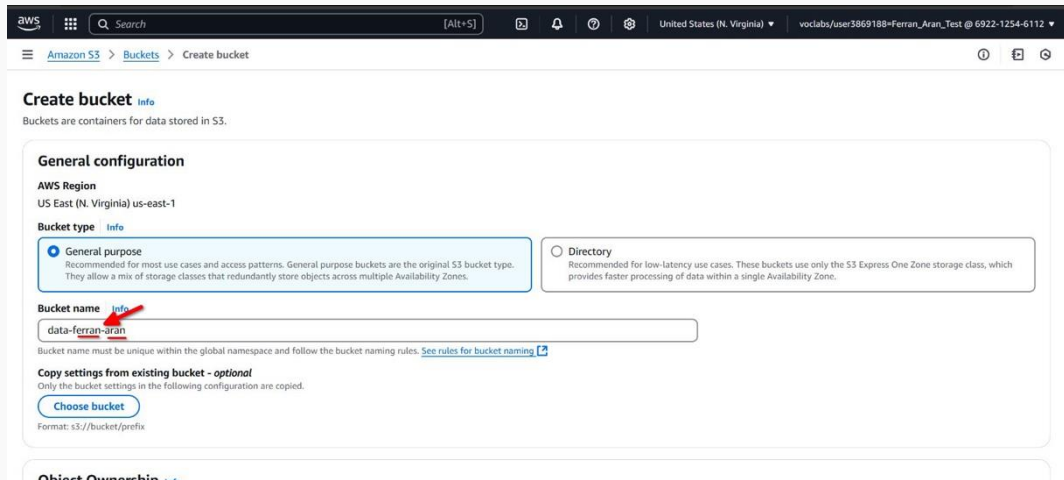
General purpose buckets (1) [Info](#) **All AWS Regions**  [Copy ARN](#) [Empty](#) [Delete](#) **Create bucket**

< 1 > 

Name	AWS Region	IAM Access Analyzer
 ferran-test123	US East (N. Virginia) us-east-1	View analyzer for us-east-1

Creating a S3 bucket

Name the bucket `data-{your-name}`. For example I will be naming it `data-ferran-aran`. **Bucket names have to be unique across all of AWS.**



aws  Search [Alt+S]     United States (N. Virginia) ▼ voclabs/user3869188+Ferran_Aran_Test @ 6922-1254-6112 ▼

Amazon S3 > Buckets > Create bucket

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [info](#)

data-ferran-aran

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [info](#)

Creating a S3 bucket

Leave everything else as default, scroll all the way down and click on **Create bucket**.

Amazon S3 > Buckets > Create bucket

ⓘ ⓘ ⓘ

Add tag

Default encryption ⓘ

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type ⓘ

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable
☒ Enable

▶ **Advanced settings**

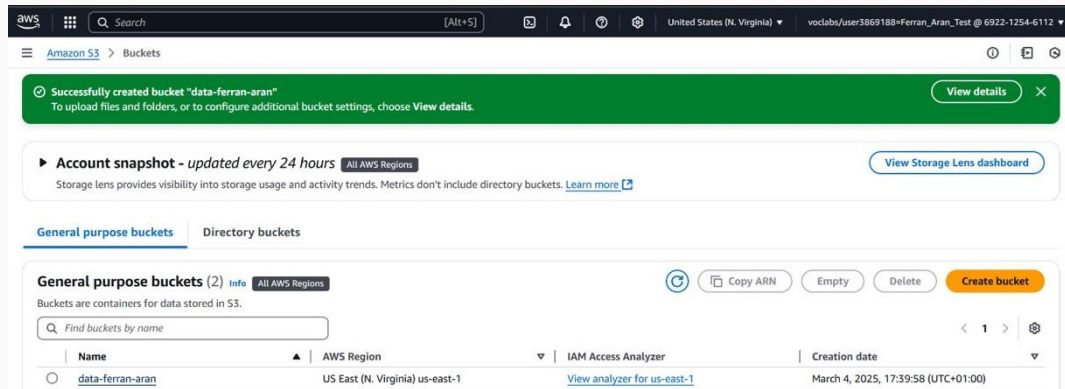
ⓘ After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel

Create bucket

Creating a S3 bucket

You will now be headed to the S3 dashboard. You should see a success message and the bucket you just created.



The screenshot shows the AWS S3 console interface. At the top, the AWS logo and navigation bar are visible, including a search bar and region selector set to 'United States (N. Virginia)'. Below the navigation bar, the breadcrumb 'Amazon S3 > Buckets' is shown. A green success banner at the top of the main content area states: 'Successfully created bucket "data-ferran-aran"'. Below this, a section for 'Account snapshot' is visible. The main content area is divided into two tabs: 'General purpose buckets' (selected) and 'Directory buckets'. Under 'General purpose buckets', there is a sub-header 'General purpose buckets (2)' and a description: 'Buckets are containers for data stored in S3.' Below the description is a search bar labeled 'Find buckets by name'. A table lists the buckets, with one bucket 'data-ferran-aran' shown. The table has columns for Name, AWS Region, IAM Access Analyzer, and Creation date. The bucket 'data-ferran-aran' is in the 'US East (N. Virginia) us-east-1' region and was created on 'March 4, 2025, 17:39:58 (UTC+01:00)'. To the right of the table, there are buttons for 'Copy ARN', 'Empty', 'Delete', and 'Create bucket'.

aws [Search] [Alt+S] [Icons] United States (N. Virginia) ▼ voclabs/user3869188=Ferran_Aran_Test @ 6922-1254-6112 ▼

Amazon S3 > Buckets

Successfully created bucket "data-ferran-aran"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours [All AWS Regions](#) [View Storage Lens dashboard](#)
Storage lens provides visibility into storage usage and activity trends. Metrics don't include directory buckets. [Learn more](#)

General purpose buckets | Directory buckets

General purpose buckets (2) [Info](#) [All AWS Regions](#)
Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	IAM Access Analyzer	Creation date
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 17:39:58 (UTC+01:00)

Copy ARN Empty Delete Create bucket

Uploading a file to the bucket

From the S3 dashboard

1. Click on the bucket.
2. Create a folder and name it *test*.
3. Click on the folder.
4. Click on Upload.
5. Click on Add files and select an example file.
6. Click Upload.

Our bucket **data-{your-name}**/ now contains one file. It is stored in **s3://data-{your-name}/test**. This file is private by default. Only the owner can access them.

From the AWS CLI

- `aws s3 cp <local-file-path> s3://<bucket-name>/<optional-target-name>`

Example (upload photo.jpg to a bucket named my-bucket):

```
aws s3 cp photo.jpg s3://my-bucket/
```

Sync with a bucket

Syncing a local directory to upload to a bucket

From the AWS CLI

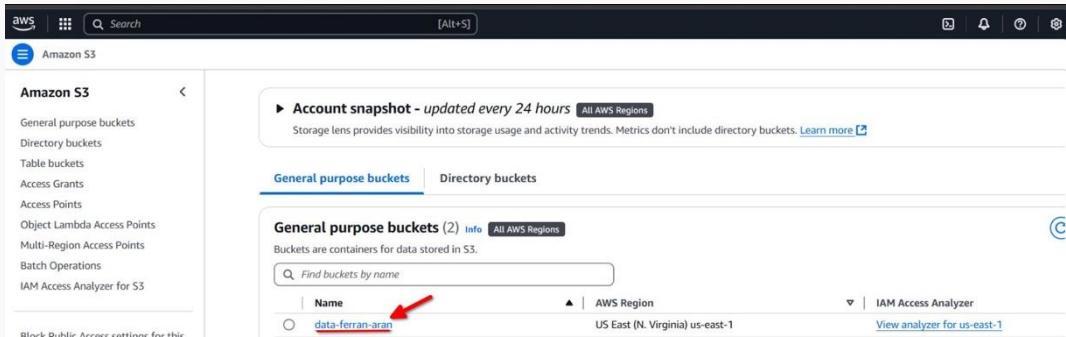
1. Go to your desktop and create a folder named `data`.
2. Create a file named `my-dataset.txt` and write some text in it. Save it.
3. Open a terminal and navigate to the folder. Use `cd Desktop`.
4. Run `aws s3 sync data s3://data-{your-name}` to upload the files to the bucket. In my case this is the result:

```
PS C:\Users\fnao\Desktop> aws s3 sync data s3://data-ferran-aran
upload: data\my-dataset.txt to s3://data-ferran-aran/my-dataset.txt
```

That's it! You have now uploaded a file to the bucket!

Inspecting the bucket from the AWS Console

Go to the S3 dashboard on AWS and click on the bucket we just created.

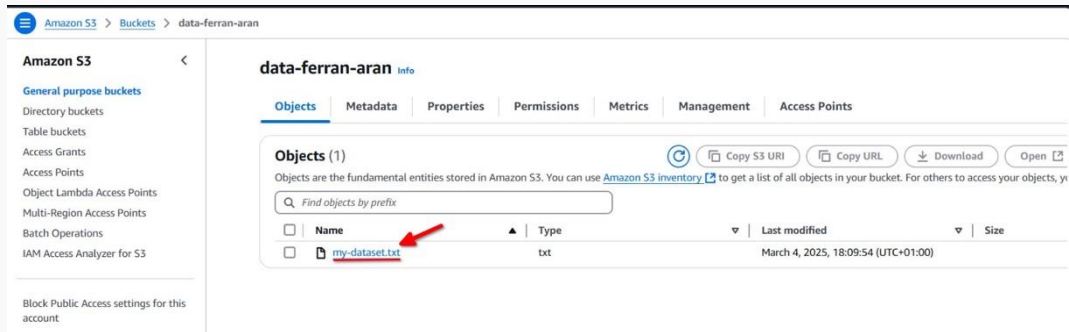


The screenshot shows the AWS S3 console interface. On the left is a navigation sidebar with the 'Amazon S3' header and a list of options: General purpose buckets, Directory buckets, Table buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, and IAM Access Analyzer for S3. The main content area has a top section for 'Account snapshot - updated every 24 hours' with a 'Learn more' link. Below this are two tabs: 'General purpose buckets' (selected) and 'Directory buckets'. The 'General purpose buckets' section shows 'Buckets are containers for data stored in S3.' and a search bar 'Find buckets by name'. A table lists the buckets with columns for Name, AWS Region, and IAM Access Analyzer. The first bucket, 'data-ferran-aran', is highlighted with a red arrow. The table also shows the region 'US East (N. Virginia) us-east-1' and a link to 'View analyzer for us-east-1'.

Name	AWS Region	IAM Access Analyzer
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1

Inspecting the bucket from the AWS Console

You will see the contents of the bucket, in this case the file `my-dataset.txt`.



The screenshot shows the AWS Amazon S3 console interface. The breadcrumb navigation at the top indicates the path: Amazon S3 > Buckets > data-ferran-aran. On the left sidebar, under 'Amazon S3', the 'General purpose buckets' section is expanded, showing various bucket types and access management options. The main content area displays the 'data-ferran-aran' bucket details. The 'Objects' tab is selected, showing a list of objects. A search bar is present above the table. The table has columns for Name, Type, Last modified, and Size. One object, 'my-dataset.txt', is listed with a type of 'txt' and a last modified date of 'March 4, 2025, 18:09:54 (UTC+01:00)'. A red arrow points to the object name 'my-dataset.txt'.

Amazon S3 > Buckets > data-ferran-aran

Amazon S3

- General purpose buckets
- Directory buckets
- Table buckets
- Access Grants
- Access Points
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

Block Public Access settings for this account

data-ferran-aran Info

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you can use [Access Points](#).

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	my-dataset.txt	txt	March 4, 2025, 18:09:54 (UTC+01:00)	

Inspecting the bucket from the AWS Console

Click on the file to see further details.

The screenshot displays the AWS Management Console interface for an Amazon S3 bucket named 'data-ferran-aran'. The breadcrumb navigation shows the path: Amazon S3 > Buckets > data-ferran-aran > my-dataset.txt. The left-hand navigation pane lists various S3 features, including 'General purpose buckets' and 'Storage Lens'. The main content area is titled 'my-dataset.txt' and includes tabs for 'Properties', 'Permissions', and 'Versions'. The 'Properties' tab is active, showing an 'Object overview' section with details such as the owner (aws-labs), region (US East (N. Virginia) us-east-1), last modified date (March 4, 2025), size (18.0 B), type (txt), and key (my-dataset.txt). On the right side, there are buttons for 'Copy S3 URI', 'Download', 'Open', and 'Object actions'. Below these buttons, the 'S3 URI' is displayed as <s3://data-ferran-aran/my-dataset.txt>, which is underlined and highlighted by a red arrow. Other details shown include the Amazon Resource Name (ARN), Entity tag (Etag), and Object URL.

Amazon S3

General purpose buckets

Directory buckets

Table buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

my-dataset.txt

Properties Permissions Versions

Object overview

Owner
aws-labs0w4907751t1669516777

AWS Region
US East (N. Virginia) us-east-1

Last modified
March 4, 2025, 18:09:54 (UTC+01:00)

Size
18.0 B

Type
txt

Key
my-dataset.txt

Copy S3 URI Download Open Object actions

S3 URI
<s3://data-ferran-aran/my-dataset.txt>

Amazon Resource Name (ARN)
<arn:aws:s3:::data-ferran-aran/my-dataset.txt>

Entity tag (Etag)
[0fee9acadfe8753f721d44b985a90c67](#)

Object URL
<https://data-ferran-aran.s3.us-east-1.amazonaws.com/my-dataset.txt>

Working with S3 from python

Configuring AWS credentials on an EC2 instance

The first thing we'll have to do is to connect to our EC2 instance we configured during last session. More information on last session can be found [here \(Session 3\)](#).

Once we are connected through SSH on a remote terminal, we'll need to configure AWS Credentials similarly to how we did on our local machine. But this time we will have to use a terminal editor.

If this steps get confusing I suggest checking [this guide](#) on the subject's website for a more detailed explanation.

Configuring AWS credentials on an EC2 instance

EC2 machines come with AWS CLI already installed so we won't have to worry about that.

Remember we first need to make sure the `.aws` folder exists in the home directory of the user we are using. If it doesn't exist we can create it by running `mkdir .aws`.

Next we need to create the `credentials` file inside the `.aws` folder. We can do this by running `nano .aws/credentials`.

The nano text editor will open and we can paste the credentials we copied from the AWS Academy website.

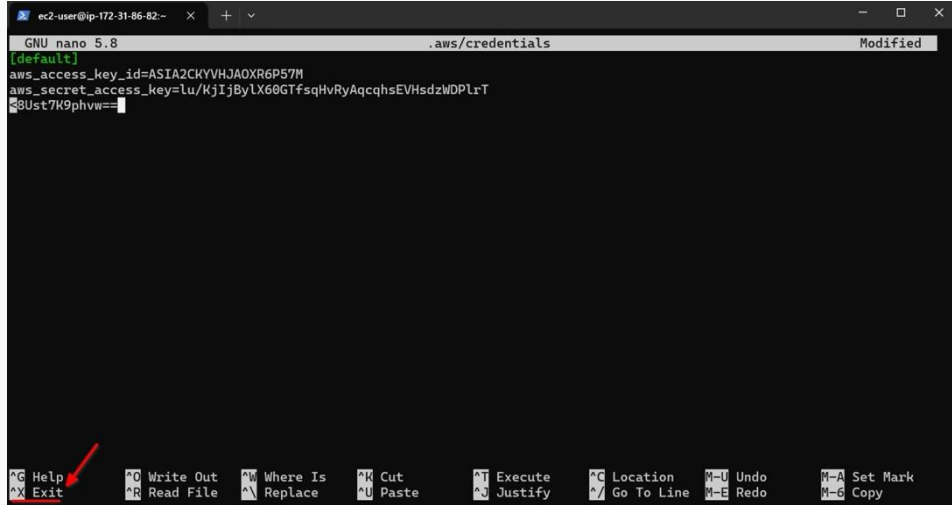
To save the file we can press `Ctrl + X` and then `Y` and finally `Enter`. See the following screenshots of the process.

Configuring AWS credentials on an EC2 instance

The image shows a terminal window with the GNU nano 5.8 text editor. The editor is open to a file named `.aws/credentials`. The terminal title bar indicates the user is `ec2-user` on a machine with IP `172-31-86-82`. The bottom status bar lists the following keyboard shortcuts:

<code>^G</code> Help	<code>^O</code> Write Out	<code>^W</code> Where Is	<code>^K</code> Cut	<code>^T</code> Execute	<code>^C</code> Location	<code>M-U</code> Undo	<code>M-A</code> Set Mark
<code>^X</code> Exit	<code>^R</code> Read File	<code>^\</code> Replace	<code>^U</code> Paste	<code>^J</code> Justify	<code>^/</code> Go To Line	<code>M-E</code> Redo	<code>M-6</code> Copy

Configuring AWS credentials on an EC2 instance



The screenshot shows a terminal window with a nano editor editing a file named `.aws/credentials`. The terminal title bar indicates the user is `ec2-user` on an instance with IP `172-31-86-82`. The nano editor's status bar shows `GNU nano 5.8` and `Modified`. The file content contains the following text:

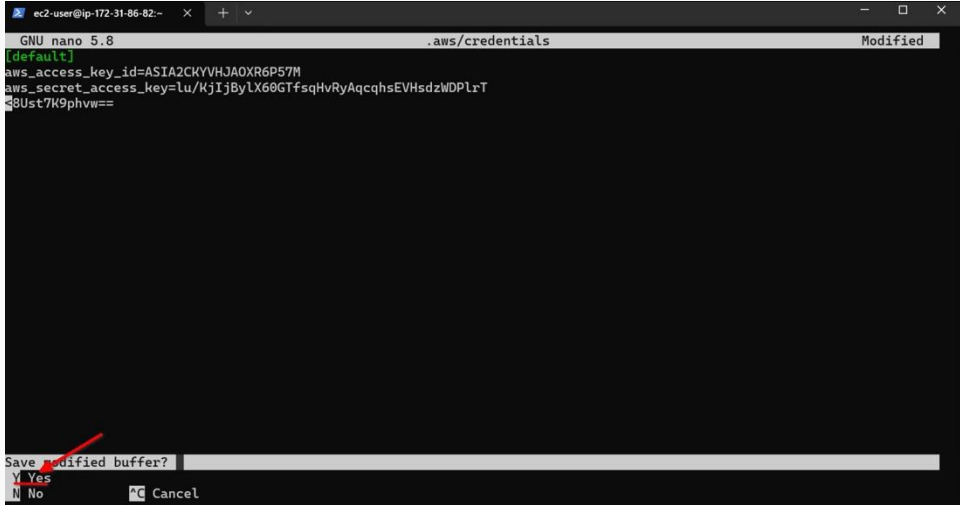
```
[default]
aws_access_key_id=ASIA2CKYVHJA0XR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
8Ust7K9phvw==
```

The cursor is positioned at the end of the last line. The bottom of the terminal displays the nano editor's help menu, which includes the following options:

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-A Set Mark
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-6 Copy

A red arrow points to the `^X Exit` option in the help menu.

Configuring AWS credentials on an EC2 instance

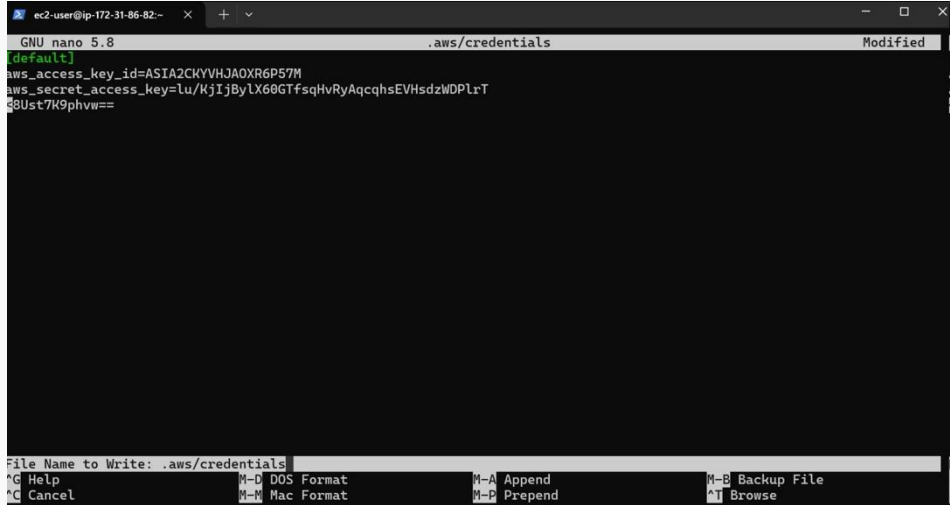


The screenshot shows a terminal window with a nano editor editing a file named `.aws/credentials`. The editor's title bar indicates it is GNU nano 5.8 and the file has been modified. The content of the file is as follows:

```
[default]
aws_access_key_id=ASIA2CKYVHJAOXR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDP1rT
8Ust7K9phvw==
```

At the bottom of the terminal, a prompt asks "Save modified buffer?". A red arrow points to the "Y" option, which is selected. The other options are "N" for No and "^C" for Cancel.

Configuring AWS credentials on an EC2 instance



The screenshot shows a terminal window with a nano editor editing a file named `.aws/credentials`. The terminal title bar indicates the user is `ec2-user` on an instance with IP `172-31-86-82`. The nano editor's status bar at the top shows `GNU nano 5.8`, the file path `.aws/credentials`, and a `Modified` status. The editor content shows the following text:

```
[default]
aws_access_key_id=ASIA2CKYVHJA0XR6P57M
aws_secret_access_key=lu/KjIjBylX60GTfsqHvRyAqcqhsEVHsdzWDPlrT
8Ust7K9phvw==
```

The bottom of the terminal shows the nano editor's command menu:

```
File Name to Write: .aws/credentials
^G Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend    ^T Browse
```

Configuring AWS credentials on an EC2 instance

We can now test if the configuration was successful by running `aws sts get-caller-identity`.

If the configuration was successful we should see something like this:

```
{  
  "UserId": "AROA2CKYVHJALK46ZMHVM:user3869188=Ferran_Aran_Test",  
  "Account": "692212546112",  
  "Arn": "arn:aws:sts::692212546112:assumed-role/voclabs/user3869188=Ferran_Aran_Test"  
}
```

Reading files from S3 with python

We are going to use the *boto3 python library* to access and write S3 files from a jupyter notebook. This library allows Python developers to write software that makes use of services like Amazon S3 on AWS.

We'll need to first activate one of the environment we created during the last session. For example, I will be using `project1` environment. Follow the steps below to activate the environment and install `boto3`:

```
cd project1
source .project1/bin/activate
pip install boto3
```

Once that is done, launch the jupyter server with the command below:

```
jupyter notebook --no-browser --port=8888 --ip=0.0.0.0
```

Remember on last session we saw the steps to access the jupyter notebook from our local machine. If you need a refresher you can check [Session 3](#) on the subject's website.

Reading files from S3 with python

Open a jupyter notebook and paste the following code:

```
import boto3

DATA_BUCKET_NAME = "data-your-name"
DATA_FILE_NAME = "my-dataset.txt" # Path to the file in S3

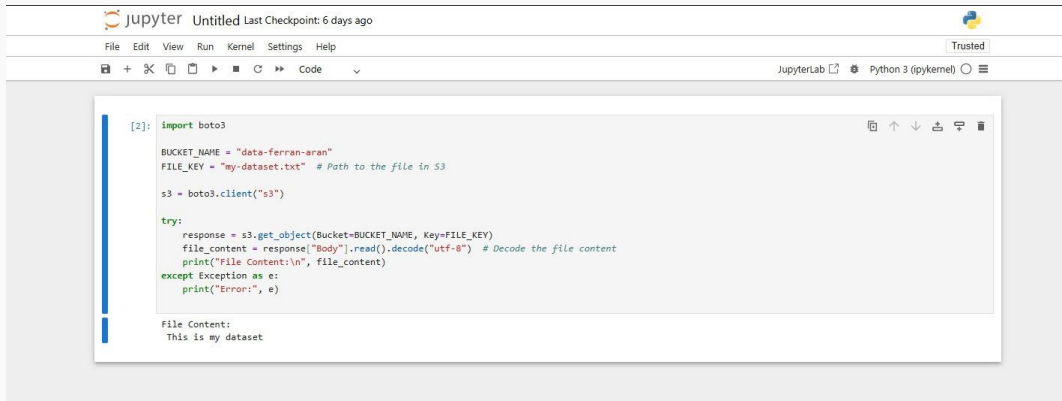
s3 = boto3.client("s3")

response = s3.get_object(Bucket=DATA_BUCKET_NAME, Key=DATA_FILE_NAME)
file_content = response["Body"].read().decode("utf-8") # Decode the file content
print("File Content:\n", file_content)
```

Be careful to replace `data-your-name` with the name of the bucket you created and `my-dataset.txt` with the name of the file you uploaded.

Reading files from S3 with python

If everything is correct you should see the content of the file printed in the notebook.



The image shows a JupyterLab interface. The top bar includes the Jupyter logo, the text "jupyter", and "Untitled Last Checkpoint: 6 days ago". On the right, there is a Python logo and a "Trusted" button. Below the top bar is a menu bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help". A toolbar with various icons is located below the menu bar. The main area contains a code cell with the following Python code:

```
[2]: import boto3

BUCKET_NAME = "data-ferran-aran"
FILE_KEY = "my-dataset.txt" # Path to the file in S3

s3 = boto3.client("s3")

try:
    response = s3.get_object(Bucket=BUCKET_NAME, Key=FILE_KEY)
    file_content = response["Body"].read().decode("utf-8") # Decode the file content
    print("File Content:\n", file_content)
except Exception as e:
    print("Error:", e)
```

Below the code cell, the output is displayed:

```
File Content:
This is my dataset
```

RECAP: Accessing the file from the notebook

To access the file from the notebook we have used our aws-credentials for the current user (owner of the bucket).

Be aware!

Someone could log into your jupyter instance in the browser and access the file using your credentials.

What to do?

In real life, we would use IAM roles to give the notebook the necessary permissions to access the file.

But, in AWS Educate, we can not use IAM roles.

Another option

Make the file public and access it without credentials :)

Writing files to S3 with python

We can also write files to S3 using boto3. Lets first create another bucket to store the files we will write from the notebook. This one we will call `results-{your-name}`.

Leave everything as default like before and scroll all the way down to click on `Create bucket`.

The screenshot shows the AWS Management Console interface for creating a new S3 bucket. The breadcrumb navigation at the top indicates the path: Amazon S3 > Buckets > Create bucket. The main heading is 'Create bucket' with an 'Info' link. Below this, a note states 'Buckets are containers for data stored in S3.' The 'General configuration' section is active, showing the 'AWS Region' as 'US East (N. Virginia) us-east-1'. Under 'Bucket type', the 'General purpose' option is selected with a radio button, while 'Directory' is unselected. The 'General purpose' description recommends it for most use cases and access patterns. The 'Bucket name' field contains 'results-ferran-aran' and is underlined in red, indicating it is required. A note below the field states that the bucket name must be unique within the global namespace and follows naming rules, with a link to 'See rules for bucket naming'. At the bottom, there is a section for 'Copy settings from existing bucket - optional', which includes a 'Choose bucket' button and a format example: 'Format: s3://bucket/prefix'.

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)
results-ferran-aran

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Writing files to S3 with python

We're now going to use the following code to write a file to the bucket we just created:

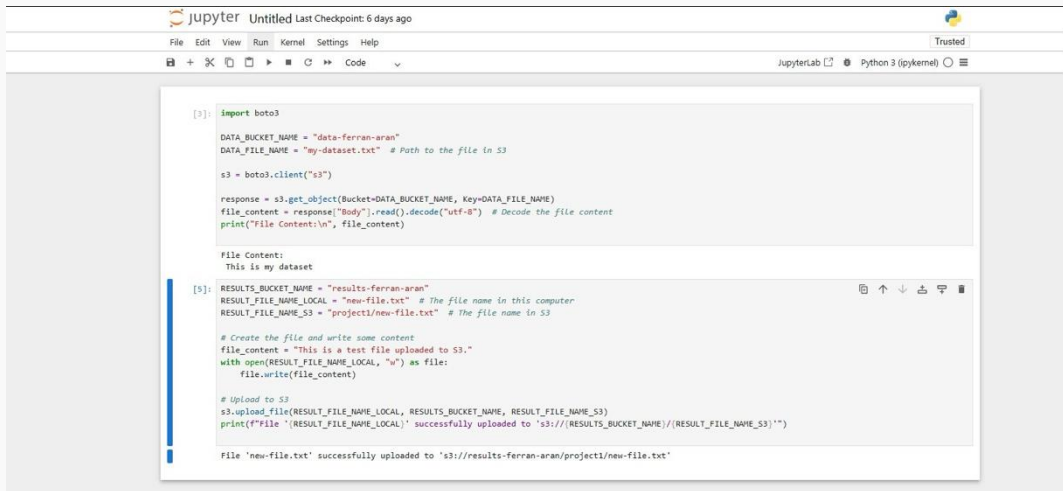
```
RESULTS_BUCKET_NAME = "results-your-name"
RESULT_FILE_NAME_LOCAL = "new-file.txt" # The file name in this computer
RESULT_FILE_NAME_S3 = "project1/new-file.txt" # The file name in S3

# Create the file and write some content
file_content = "This is a test file uploaded to S3."
with open(RESULT_FILE_NAME_LOCAL, "w") as file:
    file.write(file_content)

# Upload to S3
s3.upload_file(RESULT_FILE_NAME_LOCAL, RESULTS_BUCKET_NAME, RESULT_FILE_NAME_S3)
print(f"File '{RESULT_FILE_NAME_LOCAL}' successfully uploaded to 's3://{RESULTS_BUCKET_NAME}/{RESULT_FILE_NAME_S3}'")
```

Writing files to S3 with python

If everything is correct you should see the following:



```
[3]: import boto3

DATA_BUCKET_NAME = "data-ferran-aran"
DATA_FILE_NAME = "my-dataset.txt" # Path to the file in S3

s3 = boto3.client("s3")

response = s3.get_object(Bucket=DATA_BUCKET_NAME, Key=DATA_FILE_NAME)
file_content = response["Body"].read().decode("utf-8") # Decode the file content
print("File Content:\n", file_content)

File Content:
This is my dataset

[5]: RESULTS_BUCKET_NAME = "results-ferran-aran"
RESULT_FILE_NAME_LOCAL = "new-file.txt" # The file name in this computer
RESULT_FILE_NAME_S3 = "project1/new-file.txt" # The file name in S3

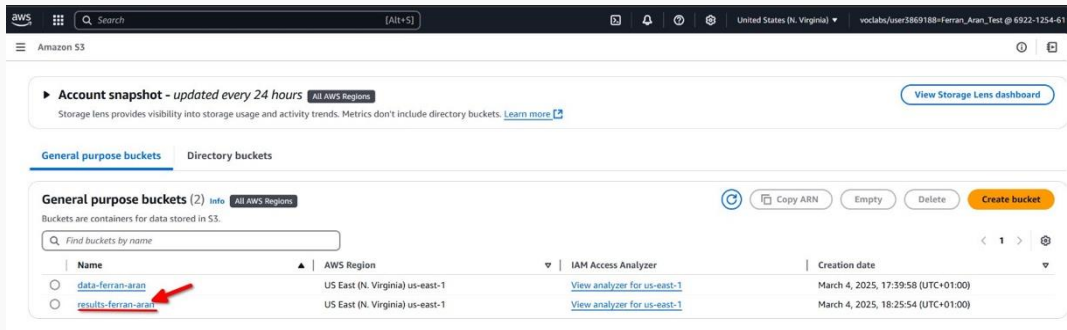
# Create the file and write some content
file_content = "This is a test file uploaded to S3."
with open(RESULT_FILE_NAME_LOCAL, "w") as file:
    file.write(file_content)

# Upload to S3
s3.upload_file(RESULT_FILE_NAME_LOCAL, RESULTS_BUCKET_NAME, RESULT_FILE_NAME_S3)
print(f"File '{RESULT_FILE_NAME_LOCAL}' successfully uploaded to 's3://{RESULTS_BUCKET_NAME}/{RESULT_FILE_NAME_S3}'")

File 'new-file.txt' successfully uploaded to 's3://results-ferran-aran/project1/new-file.txt'
```

Inspecting the bucket from the AWS Console

We can now navigate to the S3 dashboard and click on the `results-{your-name}` bucket.

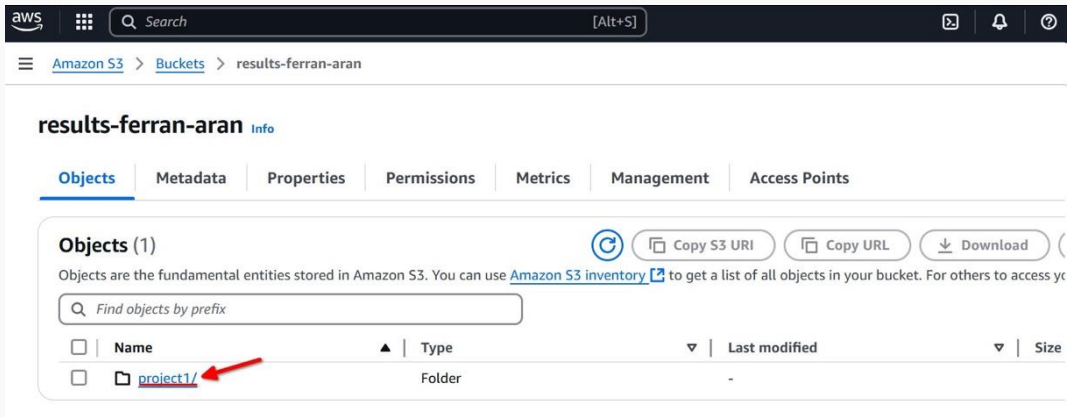


The screenshot shows the AWS S3 console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the region 'United States (N. Virginia)'. Below this, the 'Amazon S3' header is visible. The main content area shows an 'Account snapshot' and a 'View Storage Lens dashboard' button. Under the 'General purpose buckets' tab, there's a section titled 'General purpose buckets (2)' with a search bar and a table of buckets. A red arrow points to the 'results-ferran-aran' bucket in the table.

Name	AWS Region	IAM Access Analyzer	Creation date
data-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 17:39:58 (UTC+01:00)
results-ferran-aran	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 4, 2025, 18:25:54 (UTC+01:00)

Inspecting the bucket from the AWS Console

Inside the bucket we should see the folder we just created, click on it.



The screenshot shows the AWS S3 console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and a menu icon. Below the navigation bar, the breadcrumb trail reads "Amazon S3 > Buckets > results-ferran-aran". The main heading is "results-ferran-aran" with an "Info" link. Below the heading, there are tabs for "Objects", "Metadata", "Properties", "Permissions", "Metrics", "Management", and "Access Points". The "Objects" tab is selected. Under the "Objects" tab, there's a section titled "Objects (1)" with a refresh icon and three buttons: "Copy S3 URI", "Copy URL", and "Download". Below this, there's a search bar with the placeholder text "Find objects by prefix". A table lists the objects in the bucket. The table has columns for "Name", "Type", "Last modified", and "Size". The first row shows a folder named "project1/" with a folder icon, and its type is "Folder". A red arrow points to the "project1/" link in the "Name" column.

aws Search [Alt+S]

Amazon S3 > Buckets > results-ferran-aran

results-ferran-aran Info

Objects Metadata Properties Permissions Metrics Management Access Points

Objects (1) Copy S3 URI Copy URL Download

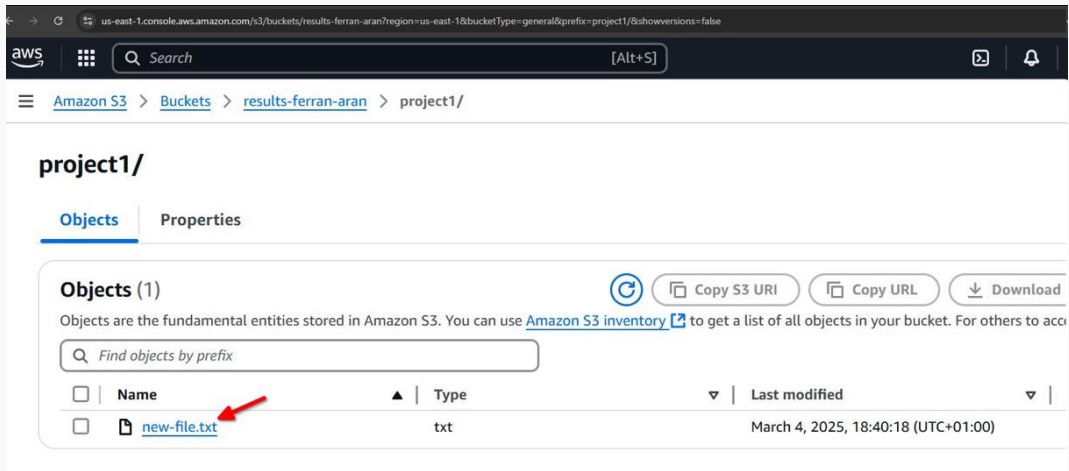
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you must grant permissions.

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	project1/	Folder	-	

Inspecting the bucket from the AWS Console

And now we should see the file we just uploaded.



The screenshot shows the AWS S3 console interface. The breadcrumb navigation at the top indicates the path: [Amazon S3](#) > [Buckets](#) > [results-ferran-aran](#) > [project1/](#). The main heading is **project1/**. Below this, there are two tabs: **Objects** (selected) and **Properties**. The **Objects (1)** section shows a list of objects. Above the list are buttons for [Refresh](#), [Copy S3 URI](#), [Copy URL](#), and [Download](#). Below these buttons is a search bar with the placeholder text "Find objects by prefix". The table below has columns for ☐ (checkbox), **Name**, **Type**, **Last modified**, and a dropdown arrow. The first row shows a file named [new-file.txt](#) with a red arrow pointing to it, a type of `txt`, and a last modified date of `March 4, 2025, 18:40:18 (UTC+01:00)`.

us-east-1.console.aws.amazon.com/s3/buckets/results-ferran-aran?region=us-east-1&bucketType=general&prefix=project1/&showversions=false

aws Search [Alt+S]

Amazon S3 > Buckets > results-ferran-aran > project1/

project1/

Objects Properties

Objects (1) Refresh Copy S3 URI Copy URL Download

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to acc

Find objects by prefix

☐	Name	Type	Last modified
☐	new-file.txt	txt	March 4, 2025, 18:40:18 (UTC+01:00)

Sync with a bucket

Syncing a local directory to download from a bucket

From the AWS CLI

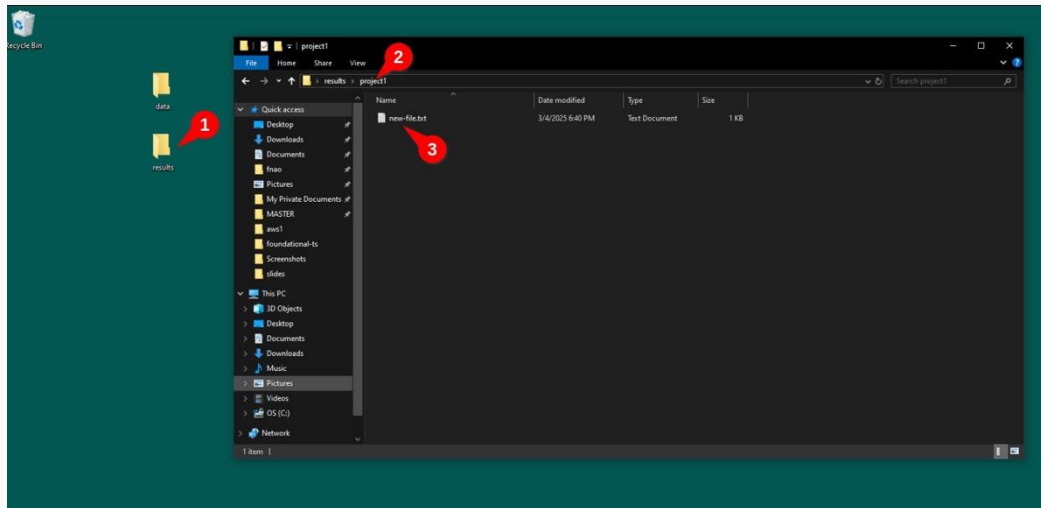
1. Create a new folder on your desktop and name it `results`.
2. Open a terminal and navigate to the folder. Use `cd Desktop`.
3. Run `aws s3 sync s3://results-{your-name} results` to download the files from the bucket.

In my case this is the result:

```
PS C:\Users\fnao\Desktop> aws s3 sync s3://results-ferran-aran results
download: s3://results-ferran-aran/project1/new-file.txt to results\project1\new-file.txt
PS C:\Users\fnao\Desktop>
```

Syncing a local directory to download from a bucket

We can use the File Explorer to navigate to the folder and see the file we just downloaded.



Recap

Recap - Today's contents

Today we have learned how to:

- Install the AWS CLI
- Configure AWS credentials
- Create an S3 bucket
- Sync a local directory to download from a bucket
- Sync a local directory to upload to a bucket
- Load files from the bucket to the notebook from python
- Write files from the notebook to the bucket from python

- With the setup we have done today we ended up with a bucket on which we can **upload datasets that we have to work on.**
- This datasets can be **accessed from any machine** with internet connection and the necessary permissions.
- We also have a results bucket which we can sync with our machine so **we have the results of our projects available locally.**

Recap - Subject's website

- Remember there is a website with useful information related to the subject.
- It has recently been updated with information about past sessions.
- Two new guides have been added to *Get started with AWS* and *Set up your Lab for every session*.